

CS471 – Lesson 21 In-Class Worksheet

Probabilistic Systems: Bayesian Inference

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Categorize variables (Query, Evidence, Hidden), compute unnormalized probabilities via factorization, apply the normalization constant α , and interpret diagnostic versus causal inference.

1. Air-Strike Mission Bayesian Network Formulation

Consider the theater air-strike planning Bayesian network:

$$W \rightarrow V, \quad W \rightarrow C, \quad V \rightarrow M, \quad C \rightarrow M$$

where W is Weather, V is Visibility, C is SATCOM, and M is Mission Outcome.

- $P(W = \text{Clear}) = 0.80, \quad P(W = \text{Severe}) = 0.20$
- $P(V = \text{Good} \mid \text{Clear}) = 0.90, \quad P(V = \text{Good} \mid \text{Severe}) = 0.30$
- $P(C = \text{Reliable} \mid \text{Clear}) = 0.85, \quad P(C = \text{Reliable} \mid \text{Severe}) = 0.40$
- $P(M = \text{Success} \mid \text{Good, Reliable}) = 0.95$
- $P(M = \text{Success} \mid \text{Good, Disrupted}) = 0.60$
- $P(M = \text{Success} \mid \text{Poor, Reliable}) = 0.50$
- $P(M = \text{Success} \mid \text{Poor, Disrupted}) = 0.10$

An operational query asks for the probability distribution of Mission Outcome (M) given that the weather is known to be Severe ($W = \text{Severe}$):

$$\mathbf{P}(M \mid W = \text{Severe})$$

a. Identify the variable partitions for this specific query:

- Query Variable (Q): _____
- Evidence Variable(s) (E): _____
- Hidden Variable(s) (H): _____

- b. Write the mathematical summation over all configurations of the hidden variables required to evaluate $P(M = \text{Success} \mid W = \text{Severe})$.

- c. Compute the numerical probability $P(M = \text{Success} \mid W = \text{Severe})$.

2. Normalization and Diagnostic Reasoning

An unnormalized diagnostic calculation yields the following joint weights for Weather (W) after an Abort is reported:

$$P(W = \text{Clear}, \text{Evidence}) = 0.0646$$

$$P(W = \text{Severe}, \text{Evidence}) = 0.0292$$

- a. Calculate the normalization constant $\alpha = \frac{1}{P(\text{Evidence})}$.

- b. Calculate the normalized posterior distribution $\mathbf{P}(W \mid \text{Evidence})$.

Part 2: Student Independent Practice

Objective: Independently execute inference by enumeration on novel multi-stage defense topologies.

Autonomous UAV Defense Network

An autonomous reconnaissance UAV operates with three binary variables in a causal chain:

$$\text{CyberJamming } (J) \rightarrow \text{SensorDegradation } (S) \rightarrow \text{TargetLoss } (L)$$

- Prior Cyber Jamming: $P(J = 1) = 0.10$, $P(J = 0) = 0.90$
- Sensor Degradation CPT:

$$P(S = 1 \mid J = 1) = 0.80, \quad P(S = 1 \mid J = 0) = 0.10$$

- Target Loss CPT:

$$P(L = 1 \mid S = 1) = 0.90, \quad P(L = 1 \mid S = 0) = 0.05$$

1. **Causal Forward Inference:** If telemetry confirms Cyber Jamming is active ($J = 1$), compute the probability that the UAV loses its target, $P(L = 1 \mid J = 1)$, by marginalizing out Sensor Degradation (S).

2. **Diagnostic Root-Cause Inference:** The mission log reports that the UAV lost its target ($L = 1$). Compute the unnormalized probability weights for $J = 1$ and $J = 0$:

a. Unnormalized weight $P(J = 1, L = 1)$:

b. Unnormalized weight $P(J = 0, L = 1)$:

- 3. Posterior Normalization:** Compute the normalization constant α and the final posterior probability that Cyber Jamming was active, $P(J = 1 \mid L = 1)$.
- 4. Tactical Interpretation:** Explain why $P(J = 1 \mid L = 1)$ is substantially lower than the sensor sensitivity $P(L = 1 \mid S = 1) = 0.90$.